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PROFESSIONAL
EXPERIENCE

SKILLS

- **Deep Learning:** PyTorch • YOLO • FasterR-CNN • ResNet • GANs • Transfer Learning
- **Computer Vision:** Object Detection • Segmentation • Pose Estimation • Face Recognition • Gaze Detection • Age/Gender Classification • People Counting • OCR • DeepSORT • MediaPipe • OpenCV
- **Tools:** Docker • GCP • Gemini API • Django • PostgreSQL • Git
- **Languages:** Python, SQL

HACKATHON EXPERIENCE

13+ hackathon projects spanning :-

- healthcare
- safety
- conservation
- education
- social impact domains

EDUCATION

Higher Secondary Education (PCB) | [2022]

Advanced Coursework:
Deep Learning, Computer Vision, ML Mathematics (self-study)

Independent Computer Vision & Deep Learning Projects| 2023–Present

- Developed 15+ computer vision applications including real-time object detection, pose estimation, face recognition, and multi-object tracking
- Deployed full-stack ML applications using Docker and Google Cloud Run with integrated databases (MySQL/ PostgreSQL)
- Specialized in YOLO-based detection systems, achieving 85%+ accuracy on custom datasets for wildlife monitoring, safety, and automation
- Built end-to-end solutions from data collection and model training through deployment and optimization

KEY PROJECTS

Helmet Detection & Number Plate Recognition System – Multi-model pipeline for automated helmet violation detection

YOLO | DeepSORT | Custom CRNN | PyTorch | OpenCV | ONNX

- Built real-time violation detection with YOLO (4 classes) and DeepSORT tracking maintaining unique IDs across video frames
- Developed custom CRNN architecture (CNN + Bi-LSTM + CTC) for number plate OCR with ONNX deployment
- Implemented 5-frame violation confirmation with automated evidence collection (annotated videos + cropped images)

MedBridge – AI Emergency Healthcare Coordination System

Gemini 2.5 Flash (Vertex AI) | Django | ElevenLabs | RAG Datadog | GCP

- Built AI emergency platform coordinating hospital selection, ambulance dispatch, and patient guidance in under 60 seconds
- Implemented RAG-based hospital matching using verified data (specialty, beds, insurance) – eliminating LLM hallucinations
- Integrated Gemini AI for 2-second symptom triage and ElevenLabs voice AI for real-time patient support during transit
- Deployed Datadog observability monitoring LLM latency, error rates, and system health for production reliability

Industrial Defect Detection using Deep Learning – Metal casting defects

PyTorch | Transfer Learning | Computer Vision | Manufacturing AI

- Built automated quality control system achieving 99.6% accuracy
- Implemented transfer learning with ResNet-18 on metal casting defects
- Zero false positives (no good parts rejected) in production validation